

Literature-Implied Answer: WLD Density Above the Continuum

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Status. *Literature-implied answer, partial subcase*, for arXiv:2301.07413. This note records a known-literature obstruction, not a new proof from this run.

Source question. Koszmider–Ryduchowski, *Equilateral and separated sets in some Hilbert generated Banach spaces*, arXiv:2301.07413, Section 5, Question, asks:

Is there a nonseparable Hilbert generated (WCG, WLD) Banach space whose unit sphere does not admit an uncountable $(1+)$ -separated set?

This is item (4) in the final question list; in the parsed source it appears at lines 962–971.

Supporting theorem. Hajek–Kania–Russo, *Separated sets and Auerbach systems in Banach spaces*, arXiv:1803.11501, Theorem A(ii), states that if X is a WLD Banach space with $\text{dens } X > \mathfrak{c}$, then the unit spheres of X and X^* contain uncountable symmetrically $(1+)$ -separated subsets. In the parsed source this theorem statement appears in the introduction around lines 147–157, with the formal item at lines 152–153.

Identification. Symmetric $(1+)$ -separation implies ordinary $(1+)$ -separation. Hence the Hajek–Kania–Russo theorem rules out WLD counterexamples to the Koszmider–Ryduchowski question whenever $\text{dens } X > \mathfrak{c}$. Since Hilbert generated spaces have weak-star dual balls that are uniform Eberlein compacta, and uniform Eberlein compacta are Corson compacta, Hilbert generated spaces are WLD; WCG spaces are WLD as well. Thus the same density obstruction applies to the Hilbert generated and WCG variants.

Scope limitations. This answers only the density-above-continuum subcase of Question 5(4). It does not settle the cases of density at most $\mathfrak{c}$, especially the ω_1 -density examples emphasized in arXiv:2301.07413. The supporting authors do not present their theorem as an answer to arXiv:2301.07413, which appeared later; the relation is an agent-identified literature implication.